

# Saman Nezafat

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## Professional Summary

Backend Software Engineer with 4+ years of experience architecting high-concurrency systems and real-time platforms serving millions of users. Proven track record in technical leadership and delivering scalable, distributed solutions in Go and Python. Open to relocation with visa sponsorship.

## Work Experience

### AryaHamrah - 2025-Present

- Stack: Go, Redis, ClickHouse, MongoDB, Docker

#### Payment Gateway

- Architected a high-concurrency payment gateway in **Go**, ensuring transaction safety via **Redis distributed locking** and **idempotency**. Implemented a modular **adapter architecture** for seamless multi-provider integration.

#### Audit Log Service

- Built a tamper-evident audit log service in **Go** with **HMAC-SHA256 hash chains** per user, a **Redis Streams** → **ClickHouse** high-throughput ingestion pipeline, and an emergency mode with disk monitoring and automatic log rotation.

#### Meter Data Management System

- Developed a **Go** backend that collects meter data and runs a multi-tier billing engine to automatically generate detailed PDF invoices.

### Top - 2023-2025

- Stack: Python, FastAPI, Go, ClickHouse, Redis, RabbitMQ, WebSocket, PostgreSQL

#### B2C Tourism Platform

- Designed and built a large-scale **tourism platform (3M+ users, 9+ B2B partners)** with **multi-gateway payment** orchestration and asynchronous, **cached** processing to reduce latency and database load.

#### Real-Time User Analytics System

- Built a high-throughput analytics system with a visual **Journey Builder** to automate personalized user flows, engagement triggers, and A/B tests; powered by **FastAPI**, **ClickHouse**, and **RabbitMQ**.

#### Messaging System

- Built a priority-managed messaging system (SMS & Push) using **Redis Streams** for efficient, real-time message queuing.

#### Real-Time Taxi POS Backend

- Developed a **WebSocket-powered** system for real-time **POS payments** and **live location streaming**.

#### AI-Powered Supermarket Agent

- Built an automated shopping assistant for WhatsApp, enabling users to place orders via voice or text, with the AI handling product search, cart management, and order creation.

### Atishahr - 2022-2023

- Stack: Python, FastAPI, PostgreSQL

## SIB Ticket System

- Optimized a key reporting endpoint, achieving a **20x performance** improvement and eliminating slowdowns under high load.
- Contributed to the project's second version using **FastAPI**, implementing **asynchronous features** to handle increased system load.

## Mehr Pars - 2021-2022

- Stack: Python, FastAPI, PostgreSQL

## Organizational Resource Management System

- Engineered a **data migration** solution, enabling seamless transitions from legacy project management systems to the Rainesh platform.

## Comprehensive Warehouse System

- Developed a **load-simulating reverse proxy** module for a large-scale warehouse system, simulating backend conditions under high load to improve frontend stability.

## Open Source

- **Muxr**, A Go WebSocket multiplexing library for efficient real-time communications.
- **IP**, A Go TCP server that returns the client's IP and country in multiple formats.
- **PriceBot**, A Go Telegram bot providing real-time currency exchange rates, gold prices, and cryptocurrency values.
- **Social Token Experiment**, A minimalist ERC-20 social token on Polygon, designed as an experiment in on-chain communication (Solidity).
- **PyBotNet**, An educational security-research framework in Python for exploring remote-control patterns via Telegram.

## Technical Skills

- **Languages**: Go, Python, Solidity
- **Frameworks & Protocols**: FastAPI, Gin, REST, WebSocket
- **Databases & Caching**: PostgreSQL, MongoDB, ClickHouse, Redis
- **Messaging & Streaming**: RabbitMQ, Redis Streams
- **Architecture**: Distributed systems, microservices, event-driven design, high concurrency, idempotency
- **DevOps**: Linux, Docker, Git, CI/CD
- **Security**: JWT, OWASP best practices

## Soft Skills

- **Leadership & Collaboration**: Led a 3-member engineering team, guiding system design and code reviews while cultivating collaboration through mentoring and documentation.

## Education

- **Bachelor of Science in Civil Engineering**

## Languages

- **English**: Professional Working Proficiency